

Estimation of Blood Glucose Level of Type-2 Diabetes Patients Using Smartphone Video through PCA-DA

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ABSTRACT

This work proposes a smart phone video based approach for the estimation of blood glucose in a non-invasive way. Videos using smartphone camera are collected from the tip of the subject's finger and the frames are subsequently converted into Photoplethysmography (PPG) waveform. Gaussian filter along with Asymmetric Least Square methods have been applied on the PPG signals to remove the high frequency noise, optical and motion interferences. Different signal features such as Systolic and Diastolic Peaks, the time difference between consecutive peaks (DelT), and First Derivative peaks have been extracted from the processed signal. Finally, Principal Component Regression (PCR) has been applied for the prediction of glucose level from the extracted features. The proposed model while applied to an unbiased dataset could predict the glucose level with a Standard Error of Prediction (SEP) of 18.31 mg/dL.

CCS CONCEPTS

- Computing methodologies~Classification and regression trees • Computer systems organization~Sensors and actuators

KEYWORDS

Photoplethysmography, PPG, Blood Glucose, Signal Processing, PCR

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1 INTRODUCTION

Diabetes mellitus is a chronic incurable disease that causes an array of serious medical complications. The glucose level of diabetes patients need to be regulated by drawing blood several times per day to measure the glucose level, and administer insulin manually [1]. Multiple components like a lancet, one-time test strips and a glucose meter are required to get the estimate. A frequent user such as Type-2 diabetes patient requires to buy and restock these components frequently. The conventional approach of blood glucose monitoring is expensive, uncomfortable and often painful.

Due to inconvenience involved with the conventional approach, the researchers had been in search for a reliable, accurate and simple method to measure the glucose level without using of selective chemical reagents or physical separations [2]. Over the last two decades, NIR spectroscopy has been established as the most popular method for noninvasive glucose monitoring [3, 4]. There are other non-invasive techniques which have not been explored enough for this purpose. Photoplethysmography also known as PPG is a low cost noninvasive technique which measures the volumetric change of blood in the arteries. PPG signal has been widely investigated for the estimation of heart rate by the researchers. Demirezen et al. [5] used Remote PPG (rPPG) to estimate heart rate from the facial videos whereas Lin et al. [6] measured the same from a wrist-acquired PPG signal under different types of motions. Banerjee et al. [7] proposed a signal processing based approach to estimate coronary artery disease (CAD) from PPG signal. Others [8, 9] have demonstrated the potential of PPG signal in analyzing sleep disorders, mental stress, emotion detection etc. However, PPG waveform which is captured in varying wavelength may contain vital information of other pathological conditions such as changes in blood glucose level. Monte-Moreno E [10] demonstrated that PPG signal collected through a pulse-oximeter can be used for the estimation of blood glucose and blood pressure with the help of Machine Learning. The process started with acquiring PPG signal from pulse-oximeter, followed by the extraction of various features and application of machine learning techniques to estimate the blood glucose and blood pressure from the signal.

Here, we propose a novel fully non-invasive technique to predict the glucose level in blood with the help of reflection PPG

imaging acquired by smartphone camera. Jonathan et al. [11] previously reported the use of mobile phone camera for reflection PPG imaging from a male subject for the estimation of heart rate. However, this is the first proof of concept demonstration for the quantitative analysis of glucose level from blood using smartphone video as of our knowledge. In our study, video data from 18 subjects have been collected from same position of the fingertips and subsequently converted into the respective waveforms. These waveforms have been preprocessed through Gaussian filter for the noise removal. Baseline wandering which is typical in this kind of data acquisition due to motion interferences has been removed using Asymmetric Least Squares (ALS) method. Different types of features have been extracted from the pre-processed PPG signal that includes the systolic and diastolic peaks, the time difference between consecutive peaks (DeIT), first derivative peaks, second derivative peaks etc. Finally, Principal Component Regression (PCR), a statistical learning approach has been applied for the estimation of glucose level. We believe our model can make blood glucose measurement easier for everyone in the near future.

2 METHODOLOGY

An overview of our proposed model has been shown in Figure 1. The process started with video data acquisition and video to PPG signal conversion. After acquiring the PPG signal, different preprocessing techniques were applied to get the clean signal. The preprocessed signal was used to extract signal features relevant to the glucose level that were then fed into the regression model. The model was trained using the feature set and reference value of

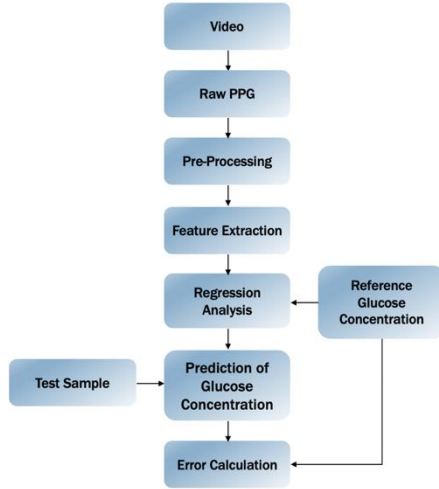


Figure 1: Proposed flow diagram

glucose taken using the commercially available glucometer (invasive method). Finally, the model was used to predict blood glucose level from unknown sample.

2.1 EXPERIMENT DETAILS

Consumer grade smartphones available in the market have good camera quality nowadays, which can be a very useful biomedical

tool thanks to the advancement of the semiconductor technology. While the quality of acquisition may vary depending on the resolution and the number of frames per second, almost all of them can be used for the capture of PPG signal. Videos for such purpose are generally captured from the parts of the body that are closest to the arteries, for example earlobe, fingertip etc. In this work, we chose fingertip as it is easier to place on the smartphone camera sensor and the subject does not require any assistance.

A commercially available smartphone (iPhone 7 plus) was used in this study. The imaging unit consists of a WLED as the illumination source next to a 12 megapixel camera at a center-to-center separation of around 5 mm. The phone supports color video recording at about 30 frames per second (fps) at a resolution of 3840 × 2160p and 60 frames per second at 1920 × 1080p. For this study, the fingertip was recorded at 30 fps at a resolution of 720p. This did not reduce signal quality as compared to that of 1080p or 4K but took less phone memory and data transfer time. A color video of the volunteer's index finger placed across both the WLED and camera sensor was recorded in MPEG-4 format onto the phones memory. During the recording the subject was requested to remain at rest to their comfortable position and not to move for as long as possible.

The MPEG-4 files were transferred from the phone memory via USB cable and iTunes software to a laptop computer for further processing. The videos were first converted to RGB JPEG frames followed by the PPG waveform extraction for red channel signals with the help of custom written Matlab (Mathworks Inc., USA) program. ROI was not chosen from the frames meaning the average intensity of the whole frame was considered. The PPG signal was extracted as:

$$ppg = MV(f, t) - MV(dc) \quad (1)$$

Here, $MV(dc)$ is the mean intensity for the entire frame, $MV(f, t)$ is the individual cell mean intensity, f is frame number and t is the time stamp of the frame.

To acquire video data, the tip of the finger was placed on the camera sensor (Figure 2). After that, a video was recorded for about 60 seconds. The duration was selected to give the data some breathing room since there is usually some movement during the placement of the fingertip on the sensor. We simultaneously collected the invasive blood sugar data using a commercial glucose meter (Accu-Check Performa and Accu-Check Active) as a reference. We processed each video by taking the red intensity value first from the whole frame, which gave us multiple values for the frame as each section of the frame had a different red intensity value. Then we averaged the values to represent that frame. After that we stored the average values in array. The array, as plotted in Figure 2, represented a PPG signal.

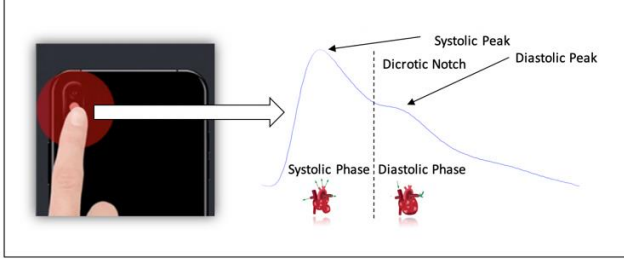


Figure 2: Basic features of a PPG signal

Video data was collected from 18 subjects aged between 15 and 61. Subject was told to place their fingertip on the camera sensor and wait for 60 seconds. Each video was captured at 30 fps on an iPhone 7 Plus. A total of 5 trials were collected from each. Alongside the video data each subject was asked to fill up some basic information such as age, state of their stomach (Full, Empty or Partially full) etc. and their invasive glucose level were measured.

2.2 PPG SIGNAL PRE-PROCESSING

For noise removal, various types of digital bandpass filters were used in the literatures. Here, the signals were cleaned with the help of Gaussian smoothing operator that works well as a bandpass filter in the frequency domain. The Gaussian distribution can be explained by the equation below where c is center and w is width:

$$G(x) = e^{-(x-c)^2/2w^2} \quad (2)$$

The degree of the smoothing can be determined by the adjustment of the standard deviation of the above function. The centroid value $c = 0.068$ and a width value $w = 0.0543$ optimized through fine-tuning within a range was used for smoothing the acquired PPG signals. Asymmetric Least Squares method (ALS), a well known for baseline correction technique [12] was chosen for the removal of motion interferences. ALS baseline correction method is very useful for correcting data with relatively narrow peaks, which is evident in Figure 3.

2.3 FEATURE EXTRACTION

Preprocessed PPG signals were subjected for feature extraction. Extracted features were Systolic and Diastolic Peaks, Delt, First Derivative peaks. Systolic Diastolic peaks are the peaks found in the preprocessed signal. DelT is basically the time difference between the systolic and diastolic peak in each cycle of the PPG signal. The processed signal was converted to its first derivative as it gives higher number of peaks increases and the peaks are also more pronounced. A 2-point central difference method was used to determine the first derivative as already described by Weng et al [13]. The number of peaks doubled from the pre-processed signal, which can be seen in Figure 3d. Here, we can see the flow of our pre-processing approach and one of the feature extracted, the first derivative peaks.

2.4 REGRESSION ANALYSIS

Regression analysis was done on raw signals, preprocessed signals, features extracted from the preprocessed signal to evaluate and

compare the performance on the prediction of glucose. In the regression analysis, a pre-defined regression method generates

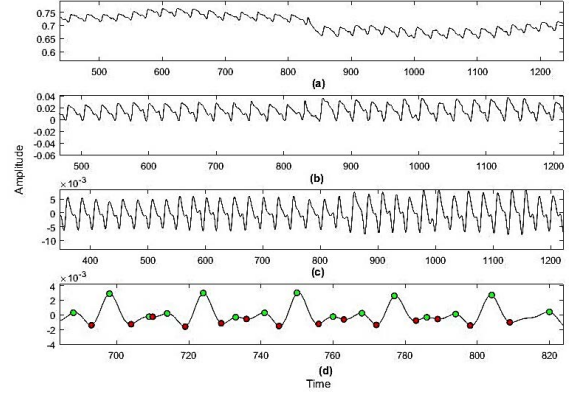


Figure 3: (a) raw PPG signal (b) preprocessed with ALS (c) preprocessed with Gaussian Filter (d) detection of first derivative peaks from the preprocessed signal

predictions from the input. There are various regression methods like classical least square regression (CLS), principle component analysis (PCA), partial least square regression (PLS), and principal component regression (PCR). PLS and PCR are widely used chemometric multivariate calibration methods and can be applied when that data set has correlated predictor variables [14]. Both regression methods create new predictor variables (components) as linear combinations of the original predictor variables. PLS creates these components while considering the observed response values. On the other hand, PCR creates component without considering the response values at all. Both regression methods have strong predictive power. PCR has been chosen for this work due to its reported ability to predict glucose level from NIR spectra with a good accuracy [15].

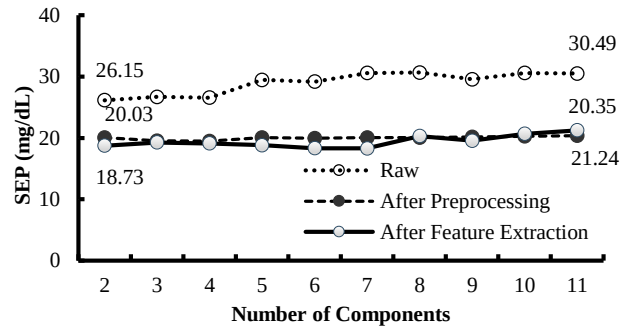


Figure 4: SEP vs. No. of Principal Components of different PCR models applied on raw PPG, preprocessed PPG, features extracted from preprocessed PPG

3 RESULTS & DISCUSSION

Different PCR models were developed with raw PPG signal (without processing), PPG signal denoised and corrected through ALS and Gaussian filter and 2nd derivative features extracted from the preprocessed PPG signal. Out of 88 trials, 75% of the trials were used for training and 25% for testing of the models. Figure 4 shows

the estimated SEP against different number of principal components (PCs) achieved for different approaches. SEP increased with the increase of No. of principal components for PCR model with raw PPG. SEP on 2nd component was 26.1571 mg/dL and on 11th component the SEP was 30.4939 mg/dL for this model. A clear reduction of SEP could be observed for all PCs for the other two models. The lowest SEP recorded for the model built with preprocessed signal was around 20 mg/dL. The SEP reduced even more (~18 mg/dL) for the PCR model built with 2nd derivative features extracted from the preprocessed PPG. The preprocessing and feature extraction have significantly improved the prediction ability of the PCR model.

After applying PCR on waveform processed through different approaches, overall better accuracy was received with 7 PCs. A comparative result of the different approaches applied while considering 7 components is given in Figure 5. It can be seen that SEP has reduced from 29.17 mg/dL to 19.96 mg/dL after applying ALS and Gaussian filter on the raw PPG signal. First derivation characteristic points extracted from the preprocessed signal produced the lowest SEP - 18.30 mg/dL when it was fed into PCR model. Other features like delT did not work well. An SEP of 21.54 mg/dL was recorded when delT was used with PCR. After regression analysis, glucose concentrations predicted through the best model (preprocessing followed by First derivation characteristics points extraction and then PCR) with lowest SEP were put into Clark Error Grid for further evaluation. Clarke error grid is a global technique used to differentiate between the glucose measurement technique under test and reference measurements [16]. From the Clarke error grid (Figure 6), It could be observed that 82.6% points are in Zone A, 17.4% points in Zone B and no

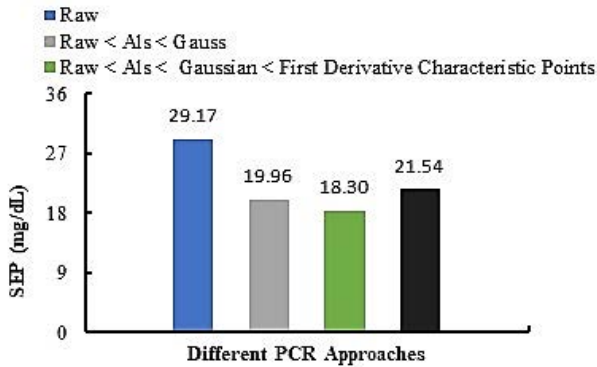


Figure 5: SEP results achieved through different preprocessing techniques applied on Raw PPG

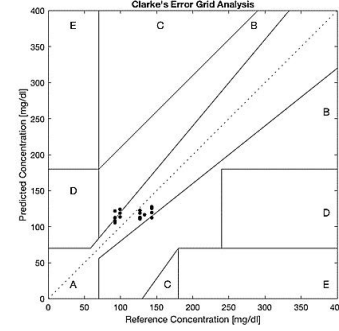


Figure 6: Clarke error grid on the predicted glucose concentration

points in the C, D, E. The predicted glucose level also fulfills the current requirement [17] for the clinically acceptable blood glucose monitoring which explains that the measurements should fall within $\pm 15 \text{ mg dL}^{-1}$ for the glucose concentration less than 75 mg dL^{-1} and $\pm 20\%$ for the concentration greater than 75 mg dL^{-1} . We believe that the PCR model built with preprocessed PPG signal followed by first derivative characteristics points have the ability to estimate glucose in clinically acceptable level.

4 CONCLUSION

Here, we have proposed a model comprised of signal processing and regression model to estimate the blood glucose level from PPG signal. A smartphone camera based PPG data acquisition and conversion method has been developed. Raw waveforms have been preprocessed and significant feature of signals have been extracted. Finally, PCR algorithm has been applied for the prediction of glucose level and the prediction ability has been validated with the commercial glucometer. The experimental results confirm the applicability of proposed method for glucose level prediction. This model can further be improved and has importance to the community as it can deliver the glucose level through smartphone available to most of the users. We in future aim at expanding the number of subjects, exploring other multivariate calibration models and implementing the model in the phone for immediate display of glucose level.

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